

Temsconsu SecureVision AI

Concealed Weapon Detection System

Technical Report

Author

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Position Applied For

Software/AI Engineer

1. Problem Understanding

Concealed weapons pose a significant security risk in public spaces such as airports, schools, event venues, government facilities, and transportation hubs. Traditional security screening methods often rely on manual inspection and human observation, which can be time-consuming and prone to error.

The objective of this project was to develop an AI-powered concealed weapon detection system capable of identifying potential weapons from thermal imagery. The system should provide real-time detection results, confidence scores, risk assessments, and actionable recommendations for security personnel.

Key objectives included:

- Detect concealed weapons from thermal images.
- Minimize false negatives to reduce security risks.
- Provide confidence-based risk assessment.
- Deliver results through an intuitive user interface.
- Create a solution that can be expanded into a real-world security screening platform.

2. Model Architecture Choice

Several candidate models were considered:

Custom CNN

A custom Convolutional Neural Network could perform image classification but would not provide object localization. While suitable for determining whether a weapon exists in an image, it cannot identify the exact location of the weapon.

MobileNet-Based Classifier

MobileNet offers lightweight deployment and efficient inference but remains primarily a classification architecture. Additional components would be required for localization.

EfficientDet

EfficientDet is a powerful object detection architecture with strong accuracy. However, it is generally more computationally intensive and slower to deploy for rapid prototyping.

YOLOv8(Selected)

YOLOv8 was selected because it offers:

- Real-time object detection
- High inference speed
- Strong detection accuracy
- Simplified deployment workflow
- State-of-the-art performance for small object detection
- Built-in training and inference capabilities through Ultralytics

YOLOv8 directly predicts object locations and confidence scores in a single forward pass, making it suitable for security screening applications where rapid detection is critical.

3. Training Process

Dataset

The dataset consisted of thermal images containing concealed weapons

Dataset Split:

- Training Images: 1,935
- Validation Images: 185
- Test Images: 91

Training Environment

- Google Colab

- Python
- Ultralytics YOLOv8
- CUDA-enabled GPU

Training Steps

1. Dataset preparation and validation.
2. Annotation verification.
3. YOLOv8 model initialization.
4. Model training on the training dataset.
5. Validation during training.
6. Best model checkpoint selection.
7. Evaluation on validation and test datasets.

After much iterations, the model was trained on 50 epochs with a batch size of 16, meaning the model will review the entire dataset 50 different times, processing the data in small groups of 16 samples at a time.

Model Output

The final trained model was exported as:

best.pt

This model was later integrated into a Streamlit-based web application for inference.

4. Evaluation Results

The trained model achieved the following metrics:

<u>Metric</u>	<u>Score</u>
Precision	0.945
Recall	0.994
mAP@50	0.986
mAP@50-95	0.629

Interpretation

Precision (94.5%)

Indicates that most detected weapons were actual weapons, resulting in relatively few false positives.

Recall (99.4%)

Demonstrates that the model successfully identified nearly all weapons present in the dataset.

mAP@50 (98.6%)

Shows excellent detection performance at an IoU threshold of 0.50.

mAP@50-95 (62.9%)

This metric evaluates localization accuracy across stricter IoU thresholds. The lower score suggests opportunities to improve bounding box precision through additional data and training optimization.

Overall Assessment

The model demonstrates strong detection capability and is suitable for proof-of-concept deployment.

5. UI/UX Design Decisions

The application was developed using Streamlit to enable rapid deployment and intuitive interaction.

Key design decisions included:

Simplicity

The interface was designed for security personnel who may not have technical expertise.

Two-Panel Visualization

Users can simultaneously view:

- Original Image
- Detection Results

This improves interpretability and trust in the system.

Risk Classification

Detection confidence is translated into:

- LOW Risk
- MEDIUM Risk
- HIGH Risk

This helps non-technical users make decisions quickly.

AI Assessment

The system generates human-readable explanations describing the detection results.

Recommended Action

The application provides security recommendations based on detection confidence.

6. Branding Implementation

To align the solution with Temsconsu's identity, the application was customized using the company's branding guidelines. It was designed to give the user a feel of a Temsconsu application, incorporating Temsconsu theme colors in the header, Logo placement and footer customization.

Implemented branding features include:

- Temsconsu logo integration
- White-themed interface
- Temsconsu blue (#0B4EA2) as primary color
- Temsconsu orange (#F26722) as accent color
- Customized footer
- Branded browser tab icon
- Professional dashboard styling

The resulting product was presented as:

SecureVision AI

A potential Temsconsu security technology solution.

7. Limitations and Next Steps

Current Limitations

Dataset Size

The training dataset is relatively small and may not represent all real-world scenarios.

Thermal Image Dependency

The model was trained primarily on thermal imagery and may require retraining for RGB camera feeds.

Single-Class Detection

The system currently detects weapons as a single category and does not differentiate between weapon types.

False Positives

Certain objects with similar thermal signatures could be incorrectly classified.

Future Enhancements

Multi-Class Detection

Detect and classify:

- Handguns
- Rifles
- Knives
- Other prohibited items

Video Stream Processing

Support live CCTV and surveillance camera feeds.

AI Agent Integration

Integrate Large Language Models to:

- Explain detections
- Generate incident reports
- Assist security operators
- Recommend response procedures

Edge Deployment

Deploy optimized models on edge devices for real-time screening.

Security Operations Dashboard

Create a centralized monitoring dashboard for large-scale deployments.

Conclusion

This project demonstrates the application of modern computer vision techniques to automated security screening. By leveraging YOLOv8, Streamlit, and Temsconsu branding, the solution provides a functional proof-of-concept for concealed weapon detection that can serve as the foundation for a scalable commercial security platform.